In [2]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [3]:
```
boston = pd.read_csv("boston.csv")
```

In [4]:
```
boston
```

Out[4]:

	TOWN	TRACT	LON	LAT	MEDV	CRIM	ZN	INDUS	CHAS	NOX	I
0	Nahant	2011	-70.9550	42.2550	24.0	0.00632	18.0	2.31	0	0.538	6.5
1	Swampscott	2021	-70.9500	42.2875	21.6	0.02731	0.0	7.07	0	0.469	6.4
2	Swampscott	2022	-70.9360	42.2830	34.7	0.02729	0.0	7.07	0	0.469	7.1
3	Marblehead	2031	-70.9280	42.2930	33.4	0.03237	0.0	2.18	0	0.458	6.9
4	Marblehead	2032	-70.9220	42.2980	36.2	0.06905	0.0	2.18	0	0.458	7.1
...	...	...	...	...	...	...	...	...	...	...	
501	Winthrop	1801	-70.9860	42.2312	22.4	0.06263	0.0	11.93	0	0.573	6.5
502	Winthrop	1802	-70.9910	42.2275	20.6	0.04527	0.0	11.93	0	0.573	6.1
503	Winthrop	1803	-70.9948	42.2260	23.9	0.06076	0.0	11.93	0	0.573	6.9
504	Winthrop	1804	-70.9875	42.2240	22.0	0.10959	0.0	11.93	0	0.573	6.7
505	Winthrop	1805	-70.9825	42.2210	19.0	0.04741	0.0	11.93	0	0.573	6.0

506 rows × 16 columns

```
The Boston Housing Dataset

The Boston Housing Dataset is a derived from information collected by the
U.S. Census Service concerning housing in the area of Boston MA. The
following describes the dataset columns:

    CRIM - per capita crime rate by town
    ZN - proportion of residential land zoned for lots over 25,000 sq.ft.
    INDUS - proportion of non-retail business acres per town.
    CHAS - Charles River dummy variable (1 if tract bounds river; 0
otherwise)
    NOX - nitric oxides concentration (parts per 10 million)
    RM - average number of rooms per dwelling
    AGE - proportion of owner-occupied units built prior to 1940
    DIS - weighted distances to five Boston employment centres
    RAD - index of accessibility to radial highways
    TAX - full-value property-tax rate per $10,000
    PTRATIO - pupil-teacher ratio by town
    B - 1000(Bk - 0.63)^2 where Bk is the proportion of blacks by town
    LSTAT - % lower status of the population
    MEDV - Median value of owner-occupied homes in $1000's
```

In [5]:
```
boston.head()
```

Out[5]:

	TOWN	TRACT	LON	LAT	MEDV	CRIM	ZN	INDUS	CHAS	NOX	RM
0	Nahant	2011	-70.955	42.2550	24.0	0.00632	18.0	2.31	0	0.538	6.575
1	Swampscott	2021	-70.950	42.2875	21.6	0.02731	0.0	7.07	0	0.469	6.421
2	Swampscott	2022	-70.936	42.2830	34.7	0.02729	0.0	7.07	0	0.469	7.185
3	Marblehead	2031	-70.928	42.2930	33.4	0.03237	0.0	2.18	0	0.458	6.998
4	Marblehead	2032	-70.922	42.2980	36.2	0.06905	0.0	2.18	0	0.458	7.147

In [6]:
```
boston.tail()
```

Out[6]:

	TOWN	TRACT	LON	LAT	MEDV	CRIM	ZN	INDUS	CHAS	NOX	RM
501	Winthrop	1801	-70.9860	42.2312	22.4	0.06263	0.0	11.93	0	0.573	6.593
502	Winthrop	1802	-70.9910	42.2275	20.6	0.04527	0.0	11.93	0	0.573	6.120
503	Winthrop	1803	-70.9948	42.2260	23.9	0.06076	0.0	11.93	0	0.573	6.976
504	Winthrop	1804	-70.9875	42.2240	22.0	0.10959	0.0	11.93	0	0.573	6.794
505	Winthrop	1805	-70.9825	42.2210	19.0	0.04741	0.0	11.93	0	0.573	6.030

In [7]:
```
boston.dtypes
```

Out[7]:
```
TOWN        object
TRACT        int64
LON        float64
LAT        float64
MEDV       float64
CRIM       float64
ZN         float64
INDUS      float64
CHAS         int64
NOX        float64
RM         float64
AGE        float64
DIS        float64
RAD          int64
TAX          int64
PTRATIO    float64
dtype: object
```

In [8]:
```
boston.columns
```

Out[8]:
```
Index(['TOWN', 'TRACT', 'LON', 'LAT', 'MEDV', 'CRIM', 'ZN', 'INDUS', 'CHA
S',
       'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX', 'PTRATIO'],
      dtype='object')
```

In [9]:
```
boston[['TOWN', 'TRACT', 'LON', 'LAT', 'MEDV', 'CRIM', 'ZN', 'INDUS', 'CHAS
       'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX', 'PTRATIO']]
```

Out[9]:

	TOWN	TRACT	LON	LAT	MEDV	CRIM	ZN	INDUS	CHAS	NOX	I
0	Nahant	2011	-70.9550	42.2550	24.0	0.00632	18.0	2.31	0	0.538	6.5
1	Swampscott	2021	-70.9500	42.2875	21.6	0.02731	0.0	7.07	0	0.469	6.4
2	Swampscott	2022	-70.9360	42.2830	34.7	0.02729	0.0	7.07	0	0.469	7.1
3	Marblehead	2031	-70.9280	42.2930	33.4	0.03237	0.0	2.18	0	0.458	6.9
4	Marblehead	2032	-70.9220	42.2980	36.2	0.06905	0.0	2.18	0	0.458	7.1
...	...	...	...	...	...	...	...	...	...	...	
501	Winthrop	1801	-70.9860	42.2312	22.4	0.06263	0.0	11.93	0	0.573	6.5
502	Winthrop	1802	-70.9910	42.2275	20.6	0.04527	0.0	11.93	0	0.573	6.1
503	Winthrop	1803	-70.9948	42.2260	23.9	0.06076	0.0	11.93	0	0.573	6.9
504	Winthrop	1804	-70.9875	42.2240	22.0	0.10959	0.0	11.93	0	0.573	6.7
505	Winthrop	1805	-70.9825	42.2210	19.0	0.04741	0.0	11.93	0	0.573	6.0

506 rows × 16 columns

In [10]:
```
# Dropping some variables
boston[['TOWN', 'MEDV', 'CRIM', 'ZN', 'INDUS', 'CHAS',
       'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX']]
```

Out[10]:

	TOWN	MEDV	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TA
0	Nahant	24.0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	29
1	Swampscott	21.6	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	24
2	Swampscott	34.7	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	24
3	Marblehead	33.4	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	22
4	Marblehead	36.2	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	22
...	...	...	...	...	...	...	...	...	...	...	...	
501	Winthrop	22.4	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	27
502	Winthrop	20.6	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	27
503	Winthrop	23.9	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	27
504	Winthrop	22.0	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	27
505	Winthrop	19.0	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	27

506 rows × 12 columns

In [11]:
```
boston["TOWN"].unique()
```

Out[11]:
```
array(['Nahant', 'Swampscott', 'Marblehead', 'Salem', 'Lynn', 'Sargus',
       'Lynnfield', 'Peabody', 'Danvers', 'Middleton', 'Topsfield',
       'Hamilton', 'Wenham', 'Beverly', 'Manchester', 'North Reading',
       'Wilmington', 'Burlington', 'Woburn', 'Reading', 'Wakefield',
       'Melrose', 'Stoneham', 'Winchester', 'Medford', 'Malden',
       'Everett', 'Somerville', 'Cambridge', 'Arlington', 'Belmont',
       'Lexington', 'Bedford', 'Lincoln', 'Concord', 'Sudbury', 'Wayland
',
       'Weston', 'Waltham', 'Watertown', 'Newton', 'Natick', 'Framingham
',
       'Ashland', 'Sherborn', 'Brookline', 'Dedham', 'Needham',
       'Wellesley', 'Dover', 'Medfield', 'Millis', 'Norfolk', 'Walpole',
       'Westwood', 'Norwood', 'Sharon', 'Canton', 'Milton', 'Quincy',
       'Braintree', 'Randolph', 'Holbrook', 'Weymouth', 'Cohasset',
       'Hull', 'Hingham', 'Rockland', 'Hanover', 'Norwell', 'Scituate',
       'Marshfield', 'Duxbury', 'Pembroke', 'Boston Allston-Brighton',
       'Boston Back Bay', 'Boston Beacon Hill', 'Boston North End',
       'Boston Charlestown', 'Boston East Boston', 'Boston South Boston',
       'Boston Downtown', 'Boston Roxbury', 'Boston Savin Hill',
       'Boston Dorchester', 'Boston Mattapan', 'Boston Forest Hills',
       'Boston West Roxbury', 'Boston Hyde Park', 'Chelsea', 'Revere',
       'Winthrop'], dtype=object)
```

In [12]:
```
boston.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 506 entries, 0 to 505
Data columns (total 16 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   TOWN     506 non-null    object
 1   TRACT    506 non-null    int64
 2   LON      506 non-null    float64
 3   LAT      506 non-null    float64
 4   MEDV     506 non-null    float64
 5   CRIM     506 non-null    float64
 6   ZN       506 non-null    float64
 7   INDUS    506 non-null    float64
 8   CHAS     506 non-null    int64
 9   NOX      506 non-null    float64
 10  RM       506 non-null    float64
 11  AGE      506 non-null    float64
 12  DIS      506 non-null    float64
 13  RAD      506 non-null    int64
 14  TAX      506 non-null    int64
 15  PTRATIO  506 non-null    float64
dtypes: float64(11), int64(4), object(1)
memory usage: 63.4+ KB
```

In [13]:
```
boston.isnull().sum()
```

Out[13]:
```
TOWN       0
TRACT      0
LON        0
LAT        0
MEDV       0
CRIM       0
ZN         0
INDUS      0
CHAS       0
NOX        0
RM         0
AGE        0
DIS        0
RAD        0
TAX        0
PTRATIO    0
dtype: int64
```

In [18]:
```
len(boston[pd.isnull(boston.CRIM)])
```

Out[18]:
```
0
```

In [14]:
```
boston.nunique()
```

Out[14]:
```
TOWN        92
TRACT      506
LON        375
LAT        376
MEDV       228
CRIM       504
ZN          26
INDUS       76
CHAS         2
NOX         81
RM         446
AGE        356
DIS        412
RAD          9
TAX         66
PTRATIO     46
dtype: int64
```

In [16]:
```
boston.shape
```

Out[16]:
```
(506, 16)
```

In [17]:
```
boston['TOWN'].value_counts()
```

Out[17]:
```
Cambridge            30
Boston Savin Hill    23
Lynn                 22
Boston Roxbury       19
Newton               18
                     ..
Medfield              1
Dover                 1
Lincoln               1
Sherborn              1
Nahant                1
Name: TOWN, Length: 92, dtype: int64
```

In [19]:
```
boston.describe()
```

Out[19]:

	TRACT	LON	LAT	MEDV	CRIM	ZN	INDUS
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000
mean	2700.355731	-71.056389	42.216440	22.528854	3.613524	11.363636	11.136779
std	1380.036830	0.075405	0.061777	9.182176	8.601545	23.322453	6.860353
min	1.000000	-71.289500	42.030000	5.000000	0.006320	0.000000	0.460000
25%	1303.250000	-71.093225	42.180775	17.025000	0.082045	0.000000	5.190000
50%	3393.500000	-71.052900	42.218100	21.200000	0.256510	0.000000	9.690000
75%	3739.750000	-71.019625	42.252250	25.000000	3.677083	12.500000	18.100000
max	5082.000000	-70.810000	42.381000	50.000000	88.976200	100.000000	27.740000

In [20]:
```
boston1 = boston[['TOWN', 'MEDV', 'CRIM', 'ZN', 'INDUS', 'CHAS',
       'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX']]
```

In [21]:
```
boston1.describe()
```

Out[21]:

	MEDV	CRIM	ZN	INDUS	CHAS	NOX	RM
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000
mean	22.528854	3.613524	11.363636	11.136779	0.069170	0.554695	6.284634
std	9.182176	8.601545	23.322453	6.860353	0.253994	0.115878	0.702617
min	5.000000	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000
25%	17.025000	0.082045	0.000000	5.190000	0.000000	0.449000	5.885500
50%	21.200000	0.256510	0.000000	9.690000	0.000000	0.538000	6.208500
75%	25.000000	3.677083	12.500000	18.100000	0.000000	0.624000	6.623500
max	50.000000	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000

In [22]:
```
boston.groupby(["TOWN"])["CRIM"].describe()
```

Out[22]:

TOWN	count	mean	std	min	25%	50%	75%	max
Arlington	7.0	0.081846	0.028149	0.05425	0.066530	0.070220	0.088125	0.13914
Ashland	2.0	0.041835	0.008987	0.03548	0.038657	0.041835	0.045012	0.04819
Bedford	2.0	0.018130	0.005289	0.01439	0.016260	0.018130	0.020000	0.02187
Belmont	8.0	0.072905	0.016492	0.05602	0.059803	0.067380	0.085068	0.10008
Beverly	6.0	0.135922	0.026892	0.10328	0.114327	0.137910	0.153168	0.17171
...	...	...	...	...	...	...	...	...
Weymouth	8.0	0.042755	0.011485	0.03041	0.033967	0.038495	0.051865	0.06151
Wilmington	3.0	0.107197	0.042860	0.05789	0.093025	0.128160	0.131850	0.13554
Winchester	5.0	0.101676	0.024731	0.06860	0.081870	0.115040	0.120830	0.12204
Winthrop	5.0	0.065132	0.026031	0.04527	0.047410	0.060760	0.062630	0.10959
Woburn	6.0	0.083835	0.015647	0.05646	0.080187	0.085470	0.093107	0.10153

92 rows × 8 columns

In [25]:
```
boston.groupby(["TOWN"])["CRIM"].describe()["mean"]
```

Out[25]:
```
TOWN
Arlington     0.081846
Ashland       0.041835
Bedford       0.018130
Belmont       0.072905
Beverly       0.135922
                ...
Weymouth      0.042755
Wilmington    0.107197
Winchester    0.101676
Winthrop      0.065132
Woburn        0.083835
Name: mean, Length: 92, dtype: float64
```

In [24]:
```
boston.groupby(["TOWN"])["CRIM"].describe()[["mean","count"]]
```

Out[24]:

TOWN	mean	count
Arlington	0.081846	7.0
Ashland	0.041835	2.0
Bedford	0.018130	2.0
Belmont	0.072905	8.0
Beverly	0.135922	6.0
...	...	...
Weymouth	0.042755	8.0
Wilmington	0.107197	3.0
Winchester	0.101676	5.0
Winthrop	0.065132	5.0
Woburn	0.083835	6.0

92 rows × 2 columns

In [27]:
```
boston.sort_values(by = "CRIM", ascending=False).head(10)
```

Out[27]:

	TOWN	TRACT	LON	LAT	MEDV	CRIM	ZN	INDUS	CHAS	NOX	R
380	Boston Charlestown	406	-71.0460	42.2300	10.4	88.9762	0.0	18.1	0	0.671	6.9(
418	Boston Roxbury	807	-71.0525	42.1989	8.8	73.5341	0.0	18.1	0	0.679	5.9
405	Boston South Boston	614	-71.0320	42.2057	5.0	67.9208	0.0	18.1	0	0.693	5.6
410	Boston Downtown	707	-71.0462	42.2075	15.0	51.1358	0.0	18.1	0	0.597	5.7
414	Boston Roxbury	801	-71.0420	42.1985	7.0	45.7461	0.0	18.1	0	0.693	4.5
404	Boston South Boston	613	-71.0345	42.2001	8.5	41.5292	0.0	18.1	0	0.693	5.5
398	Boston South Boston	606	-71.0230	42.2090	5.0	38.3518	0.0	18.1	0	0.693	5.4
427	Boston Roxbury	816	-71.0542	42.1961	10.9	37.6619	0.0	18.1	0	0.679	6.2(
413	Boston Downtown	710	-71.0451	42.2015	16.3	28.6558	0.0	18.1	0	0.597	5.1
417	Boston Roxbury	805	-71.0498	42.2007	10.4	25.9406	0.0	18.1	0	0.679	5.3(

In [ ]:
```

```